



CF-Loss: Clinically-relevant feature optimised loss function for retinal multi-class vessel segmentation and vascular feature measurement

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ABSTRACT

Characterising clinically-relevant vascular features, such as vessel density and fractal dimension, can benefit biomarker discovery and disease diagnosis for both ophthalmic and systemic diseases. In this work, we explicitly encode vascular features into an end-to-end loss function for multi-class vessel segmentation, categorising pixels into artery, vein, uncertain pixels, and background. This clinically-relevant feature optimised loss function (CF-Loss) regulates networks to segment accurate multi-class vessel maps that produce precise vascular features. Our experiments first verify that CF-Loss significantly improves both multi-class vessel segmentation and vascular feature estimation, with two standard segmentation networks, on three publicly available datasets. We reveal that pixel-based segmentation performance is not always positively correlated with accuracy of vascular features, thus highlighting the importance of optimising vascular features directly via CF-Loss. Finally, we show that improved vascular features from CF-Loss, as biomarkers, can yield quantitative improvements in the prediction of ischaemic stroke, a real-world clinical downstream task. The code is available at <https://github.com/rmapoh/feature-loss>.

1. Introduction

The significance of retinal vasculature for assessing ophthalmic disease has been well studied, such as venous beading as a hallmark for diagnosing diabetic retinopathy. Characterising vasculature with clinically-relevant vascular features can further provide valuable insights into systemic disease (Wagner et al., 2020; Cheung et al., 2021; Wong and Mitchell, 2004), a field which has been termed ‘oculomics’ (Wagner et al., 2020). For example, increased venular tortuosity is associated with hypertension and increased body-mass index (Owen et al., 2019; Cheung et al., 2011). Eye screening provides a non-invasive and economical observation into the microvasculature and central nervous system. Retinal multi-class vessel segmentation followed by a wide range of clinical feature estimation is considered a standard pipeline to explore disease markers and monitor the progression of ophthalmic and systemic diseases (Cheung et al., 2011; De Fauw et al., 2018; Seidelmann et al., 2016), as shown in Fig. 1.

The last two decades have produced a variety of software tools designed to measure the vascular features for clinical applications and

research (Wong et al., 2006; Perez-Rovira et al., 2011; Fraz et al., 2015; Shi et al., 2022; Zhou et al., 2022). Singapore I Vessel Assessment (SIVA) (Wong et al., 2006) was used to prove that larger retinal venular calibre is associated with risk of stroke in the Singapore Malay Eye Study. Vessel Assessment and Measurement Platform for Images of the RETina (VAMPIRE) (Perez-Rovira et al., 2011) verified that lower fractal dimension was associated with increased odds of albuminuria in the UK Biobank cohort study. Other recent softwares, such as the Quantitative Analysis of Retinal Vessel Topology and size (QUARTZ), focus on translating advanced segmentation methodologies for accurate feature measurement (Fraz et al., 2015; Zhou et al., 2022; Shi et al., 2022). For example, the open-source Automated Retinal Vascular Morphology Quantification (AutoMorph) used segmentation methods which achieved state-of-the-art performance in multi-class vessel segmentation (Zhou et al., 2022). All these software tools (SIVA, VAMPIRE, QUARTZ, AutoMorph, etc.) measure vascular features based

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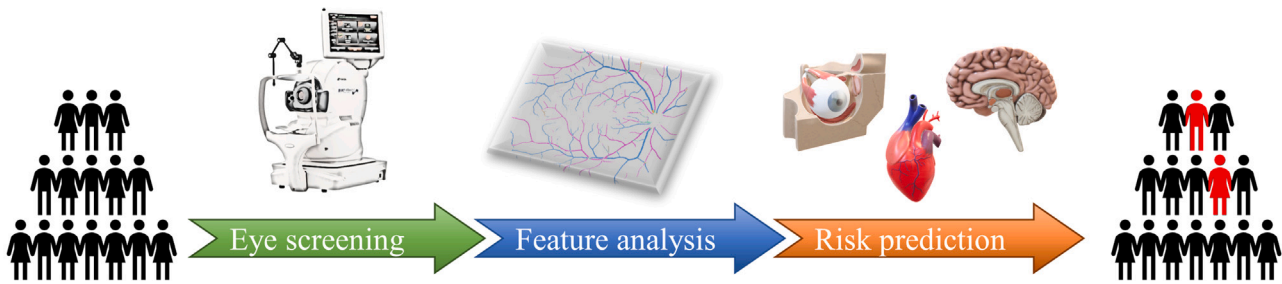


Fig. 1. Pipeline of employing retinal vascular features for ophthalmic and systemic disease monitoring, including risk prediction of cardiovascular and neurodegenerative diseases. After successive operations of eye screening, feature analysis, and risk prediction, the individuals at high risk of ocular and systemic diseases (highlighted in red) are identified from a large cohort. Feature analysis includes multi-class vessel segmentation and vascular feature measurement, which substantially affect reliability and robustness.

on retinal multi-class vessel maps. Empirically, an accurate segmentation map is essential to obtain precise features. To achieve accurate vascular features, substantial recent efforts have focused on multi-class vessel segmentation (Mookiah et al., 2021; Chen et al., 2021a; You et al., 2022).

Traditional multi-class vessel segmentation involves two steps — vessel segmentation and artery/vein classification. Two main categories of methods, namely density feature-based (Huang et al., 2018; Mirsharif et al., 2013; Niemeijer et al., 2009; Xu et al., 2017; Niemeijer et al., 2011; Zamperini et al., 2012) and graph-based (Dashtbozorg et al., 2013; Estrada et al., 2015; Srinidhi et al., 2019; Xie et al., 2020; Zhao et al., 2019), segment vessels from the retinal fundus photographs and classify the segmented pixels as arteries and veins, respectively with hand-crafted features and topological knowledge. Specifically, feature-based methods extract the hand-crafted features from image pixels and use clustering techniques, such as K-nearest neighbours (KNN) algorithm (Niemeijer et al., 2009; Xu et al., 2017), on these features to classify artery and vein. Graph-based methods include both feature information and topology knowledge to classify artery and vein.

Recently, the development of deep learning models has further boosted multi-class vessel segmentation performance enabled by the powerful capability of representation learning (Hemelings et al., 2019; Galdan et al., 2019; Zhou et al., 2021). These deep learning-based methods segment arteries and veins directly from retinal fundus photographs using an end-to-end network. Welikala et al. (2017) combined convolution layers and fully connected layers for the first deep learning-based artery/vein segmentation. After that, more complex networks, such as U-net (Ronneberger et al., 2015) and its variants (Xu et al., 2018; Li et al., 2019; Galdan et al., 2019, 2022), have been employed in multi-class segmentation to achieve better segmentation performance. Recent methods further improve multi-class vessel segmentation by incorporating extra constraints in network training (Hu et al., 2019; Li et al., 2020; Zheng et al., 2021; Shit et al., 2021; Chen et al., 2021b, 2020; Zhou et al., 2021). One exemplar is that some researchers (Hu et al., 2019; Li et al., 2020; Chen et al., 2021b) devise topology-preserved loss functions for network training. This regulates the networks to segment vessels with correct topology, which can be used to alleviate the issue of vessel fragments, a well identified vessel segmentation challenge.

Previous work assumes that improving pixel-based segmentation metrics leads to precise vascular features, thus focusing on raising pixel-based metrics while ignoring the vascular feature quantification which directly contributes to biomarker discovery and disease diagnosis (Hemelings et al., 2019; Galdan et al., 2019; Zhou et al., 2021; Mookiah et al., 2021). Additionally, vascular features, as clinically meaningful characteristics of vasculature, can provide constraints to regulate multi-class vessel segmentation, but have not been exploited yet. Here, we propose a clinically-relevant feature optimised loss function (CF-Loss) to simultaneously optimise multi-class vessel segmentation and vascular feature measurement. The key advance is to incorporate in the training loss a term that explicitly optimises the ability to generate downstream clinical features (vessel density and fractal

dimension) from the segmented vessels. We hypothesise that the end-to-end approach we propose both reduces errors in clinically-relevant features and enhances the underlying multi-class segmentation, avoiding the limitations of optimising simple pixel-based performance metrics. We demonstrate its potential using three public datasets and multiple network architectures. We summarise the contribution as below:

- To our knowledge, this is the first work that explicitly encodes clinically-relevant vascular features as an end-to-end loss function. CF-Loss works as an extra constraint for network training, being complementary to pixel-based segmentation loss.
- We have verified the efficacy of CF-Loss in multi-class vessel segmentation, with two standard segmentation networks, U-Net and BF-Net, on three public datasets — the DRIVE-AV, LES-AV, and HRF-AV datasets.
- We report that vessel maps with high pixel-based segmentation metrics do not necessarily derive accurate vascular features, from three aspects: network backbone, segmentation category, and loss function, highlighting the importance of explicitly optimising vascular features with CF-Loss for downstream clinical tasks.
- We are the first to quantitatively verify the clinical downstream contribution brought by improving segmentation and feature measurement in retinal imaging. CF-Loss improves prediction of ischaemic stroke, a real-world downstream clinical task.

2. Related work

2.1. Artery/vein classification and segmentation

Artery/vein segmentation methods can be divided into two main categories, traditional methods and deep learning-based methods. Traditional methods first segment vessel pixels and then classify the pixels as arteries or veins. Depending on whether the classifiers are developed with artery/vein labels, traditional methods can be further categorised as supervised and unsupervised methods (Mookiah et al., 2021).

Supervised methods classify the artery and vein with trained classifiers, such as linear discriminant analysis (LDA) (Huang et al., 2018; Mirsharif et al., 2013; Niemeijer et al., 2011; Dashtbozorg et al., 2013), k-nearest neighbour (kNN) (Xu et al., 2017; Niemeijer et al., 2009), and random forest (Srinidhi et al., 2019). Mirsharif et al. (2013) investigated the variance and difference of pixel intensity of vessel walls and centrelines, as the input to linear discriminant analysis classifier. Huang et al. (2018) collected 455 features for vessel centreline pixels and applied a generic search to choose informative features for artery/vein classification. Dashtbozorg et al. (2013) separated the segments between intersections as sub-graphs and classified each sub-graph as artery or vein according to intensity features. Xu et al. (2017) added the first-order and second-order texture features for the kNN classifier. Srinidhi et al. (2019) exploited local information to disentangle the whole vessel map into multiple subtrees and used the random forest to

classify these subtrees into arteries and veins. Zamperini et al. (2012) studied 16 mixing features, including colour contrast inside and outside the vessels and spatial information, for artery/vein classification with a linear Bayes normal classifier. Estrada et al. (2015) incorporated global domain-specific features and a heuristic optimisation algorithm into the tree topology estimation framework to build a likelihood model for artery/vein classification. Unsupervised methods (Zhao et al., 2019; Relan et al., 2019; Joshi et al., 2014) mainly use the clustering technique and Gaussian mixture model to classify the arteries and veins. Zhao et al. (2019) employed an active contour model (Zhao et al., 2015) to segment the vessels and formalised the artery/vein classification as a pairwise clustering problem, which split the entire vessel graph as subtrees and classified artery/vein via dominant set clustering. Joshi et al. (2014) extracted the vessel colour properties and classified the arteries and veins with a fuzzy C-means clustering algorithm. Relan et al. (2019) used orthogonal locality preserving projections to find the most dominant vessel features and classified the arteries and veins with a Gaussian Mixture Model with Expectation–Maximisation.

Deep learning-based methods have been widely explored for artery/vein segmentation (Hemelings et al., 2019; Galdran et al., 2019; Zhou et al., 2021). Unlike traditional feature-based and graph-based methods, deep learning-based methods can produce multi-class segmentation map in an end-to-end manner. Welikala et al. (2017) used six layers, including three convolutional layers and three fully connected layers, to segment arteries and veins from image patches. Xu et al. (2018) configured a fully convolutional network and a domain-specific loss function to improve the overall performance. Li et al. (2019) combined U-net and multi-scale deep supervision for extracting artery/vein details. Galdran et al. (2019) revealed that uncertain pixels, e.g. pixels at vessel intersection, existed due to the limitation of 2D images. Galdran et al. (2022) devised a coarse-to-fine network which consists of two simplified U-net to extract the artery and vein. Zhou et al. (2021) proposed to correct segmentation errors at the vessel intersections by information fusion between multi network branches.

All these methods focus on improving multi-class vessel segmentation by optimising pixel-based loss functions, which rely on the intuitive assumption that higher pixel-based performance metrics correspond to more precise vascular features. In our study, we test the assumption and explicitly encode the calculation of vascular features in end-to-end loss functions, thus enhancing both multi-class vessel segmentation and vascular feature measurement.

2.2. Topological constraints in neural networks

Although deep learning-based methods achieve good performance in artery/vein segmentation, artery/vein connectivity is challenging to maintain with networks optimised by pixel-based segmentation loss functions. To this end, some recent work has focused on encoding topological knowledge in networks. The most common way is to construct a topological constraint for network training, thus learning to segment the artery and vein with good connectivity and topology (Chen et al., 2019; Shit et al., 2021; Zheng et al., 2021; Hu et al., 2019; Li et al., 2020; Chen et al., 2020, 2021b). Let $S, T \in [0, 1]$ denote a segmentation map and a ground-truth map respectively. Chen et al. (2019) designed a loss combining the length and region information to find an active contour which is a global minimisation of active contour energy, i.e. $\sum \sqrt{(\nabla S_x)^2 + (\nabla S_y)^2} + \epsilon + \sum (S(1-T)^2 + (1-S)T^2)$ where ∇S_x and ∇S_y indicate the pixel gradients in x and y axes respectively. Zheng et al. (2021) reformulated the graph cuts loss function which focused on a dual penalty to optimise the regional properties (binary cross entropy between S and T) and boundary regularisation $\sum \{1 - |S_u - S_v|\} \cdot \delta(T_u, T_v)$, where (u, v) represents adjacent pixel pair and $\delta(T_u, T_v) = 1$ if $T_u \neq T_v$. Shit et al. (2021) proposed centreline-based Dice, $clDice = Dice(ske(S), ske(T))$ where $ske(\cdot)$ skeletonises the vessels to centrelines and $Dice(\cdot)$ calculates Dice score, to evaluate the accuracy

of segmented centreline and theoretically proved that $clDice$ guarantees topology preservation up to homotopy equivalence for segmentation.

Inspired by the constraints based on topology, we propose new regularisation with vascular features to improve biomarker quantification, which directly contributes to downstream clinical tasks. Moreover, a diverse range of vascular features supplements additional constraints absent with topological constraints, such as vessel density and fractal dimension.

2.3. Clinically-relevant vascular features

A wide range of vascular features, including fractal dimension and vessel density, have been proven as potential biomarkers for ocular diseases and systemic diseases.

Fractal dimension characterises vasculature shape complexity. Liew et al. (2011) found the lowest and highest quartiles of retinal vascular fractal dimension had higher 14-year coronary heart disease mortality. Liew et al. (2008) discovered that fractal dimension was statistically significantly lower in participants with than without hypertension. Cheung et al. (2009) recognised that the greater retinal fractal dimension was independently associated with a larger odd of early diabetic retinopathy in type 1 diabetes, which indicated that fractal dimension may allow quantitative measurement of early diabetic microvascular damage. Shi et al. (2020) found that the retinal capillary complexity in the inferior quadrant was negatively correlated with the visual acuity and Parkinson disease duration.

Vessel density quantifies the ratio of vessel pixels to the whole image. Chua et al. (2019) found that sparser vessel density was associated with higher systolic blood pressure. Chang et al. (2019) recognised that reduced vessel density was linked with age and longer diabetes duration in the African American Eye Disease Study. Yu et al. (2017) observed that retinal vessel density decreased with greater severity of obstructive sleep apnoea–hypopnoea syndrome. Akil et al. (2017) found a significant difference in vessel density between normal eyes and those with early glaucoma.

These studies all show the great potential of vascular features in clinical research. What motivates our work is robust automated estimation of these features, which current image analysis algorithms do not optimise. These vascular features are calculated in standard formulas and have not been optimised in end-to-end networks due to some non-differentiable operations. In our work, we propose approximations and validate the performance with multiple datasets and network architectures.

3. Methods

We explicitly encode the vascular features as end-to-end loss functions, as shown in Fig. 2. Among a wide range of informative vascular features (Zhou et al., 2022; Wagner et al., 2020), we focus on two specific features: vessel density and fractal dimension (box count), which we favour for their computational efficiency. However, the concept extends to other image-based vascular features.

We denote $S, T \in [0, 1]^{h \times w}$ as a segmentation map and a ground-truth map respectively, where h and w indicate the height and width of the image.

3.1. Vessel density loss function

Vessel density function v measures a ratio of vessel area A to the whole image area, e.g. $v(S) = \frac{A(S>0)}{h \times w}$ where $S > 0$ selects the potential vessel area. The vessel density loss $Loss_V(S, T)$ can be expressed as

$$\begin{aligned} Loss_V(S, T) &= \left| \frac{A(S > 0)}{h \times w} - \frac{A(T > 0)}{h \times w} \right| \\ &= \left| \frac{\sum S}{h \times w} - \frac{\sum T}{h \times w} \right|. \end{aligned} \quad (1)$$

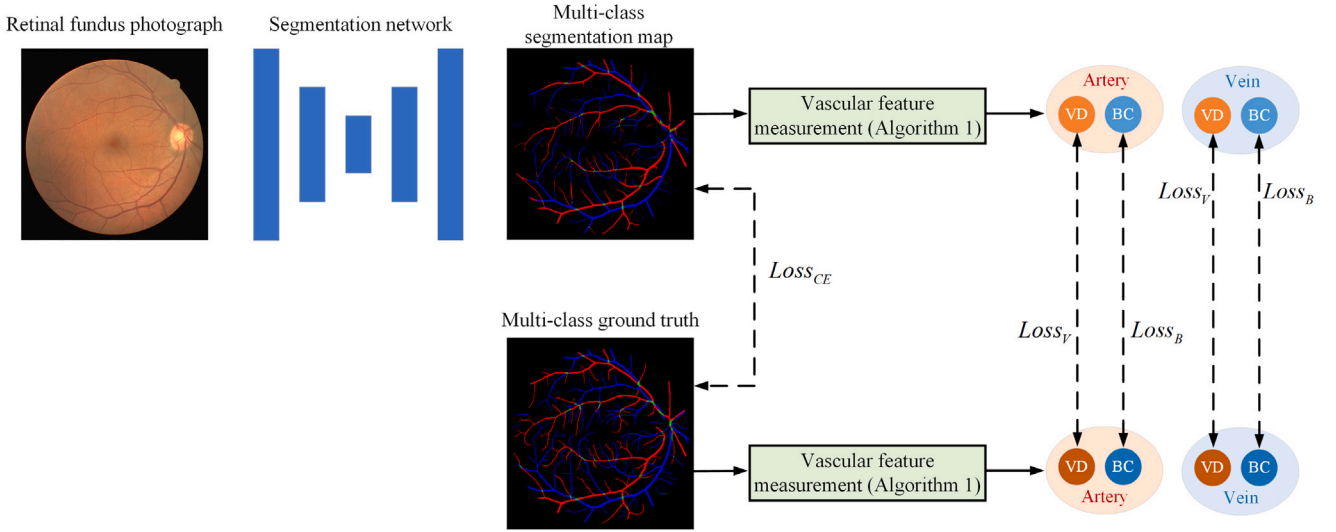


Fig. 2. Schematic of the proposed loss function CF-Loss. We measure two vascular features with Algorithm 1 and encode them into loss functions, the vessel density loss $Loss_V$ (Eq. (2)) and box count loss $Loss_B$ (Eq. (6)), to regulate the segmentation network to accurately segment vessel maps specifically to derive accurate fractal dimension and vessel density.

Algorithm 1 Vessel density and soft box count

Input: Segmentation map $S \in \mathbb{R}^{h \times w}$

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1:  $v(S) = \sum_{i=1}^{i=h} \sum_{j=1}^{j=w} S_{i,j} / (h \times w)$   $\triangleright$  calculate vessel density
2:  $\epsilon = \{2^i | i \in \mathbb{Z}, 2 \leq 2^i \leq \min\{h, w\}\}$   $\triangleright$  a set of box sizes
3:  $N(S, \epsilon), \hat{N}(S, \epsilon), \tilde{N}(S, \epsilon) = [], []$   $\triangleright$  box count list
4: for  $k \leftarrow \epsilon$  do  $\triangleright$  iterate box size
5:    $S' = \text{zeropad}(0, k - h\%k, 0, k - w\%k)(S)$ 
6:    $S' = \text{averpool}(\text{kernel size} = k, \text{stride} = k)(S')$ 
7:    $c = S' > 0$   $\triangleright$  Index of boxes including vessels
8:    $N(S, \epsilon).append(\sum_{i=1}^{i=h/k} \sum_{j=1}^{j=w/k} (\mathbb{1}_c)_{i,j})$   $\triangleright$  raw box count
9:    $\hat{N}(S, \epsilon).append(\sum_{i=1}^{i=h/k} \sum_{j=1}^{j=w/k} (S'_c)_{i,j})$   $\triangleright$  first soft box count
10:   $\tilde{N}(S, \epsilon).append((S'_c)_{i,j})$   $\triangleright$  second soft box count
11: end for
12: return  $v(S), N(S, \epsilon), \hat{N}(S, \epsilon), \tilde{N}(S, \epsilon)$ 

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For multi-class segmentation task, we get a multi-class segmentation map with four channels $S_m = (S_b, S_a, S_v, S_u)$, which respectively correspond to probability map of background S_b , artery S_a , vein S_v , and uncertain pixels S_u . We calculate the vessel density loss function for artery and vein for network training

$$Loss_V(S_m, T_m) = \left| \frac{S_a - T_a}{h \times w} \right| + \left| \frac{S_v - T_v}{h \times w} \right|, \quad (2)$$

where T_a and T_v indicate the ground truth of artery and vein.

3.2. Box count loss function

Fractal dimension function f evaluates the vessel morphology complexity. Minkowski-Bouligand dimension (also known as box-counting dimension) is commonly used in determining fractal dimension and characterising retinal vasculature (Falconer, 2004). For vessel map T , the fractal dimension is

$$f(T) = \lim_{\epsilon \rightarrow 0} \frac{\log N(T, \epsilon)}{\log(1/\epsilon)}, \quad (3)$$

where ϵ indicates square box size and $N(T, \epsilon)$ represents the number of boxes required to cover segmented vessels in T . ϵ needs infinitely close to 0 in Eq. (3).

In practice, we choose a set of box sizes $\epsilon = \{2^i | i \in \mathbb{Z}, 2 \leq 2^i \leq \min\{h, w\}\}$ as x axis, and $N(T, \epsilon)$ as y axis. Least-square

regression is used to fit a straight line with slope approximating $f(T)$. We denote error on the y axis

$$\Delta y = \log(N(T, \epsilon)) - \log(N(S, \epsilon)) = \log \frac{N(T, \epsilon)}{N(S, \epsilon)}. \quad (4)$$

When ϵ is small, $N(T, \epsilon) \approx \frac{\text{area}(S > 0)}{\epsilon^2}$, thus Eq. (4) can be transferred as

$$\Delta y = \log \frac{N(T, \epsilon)}{N(S, \epsilon)} \approx \log \frac{\text{area}(T > 0)/\epsilon^2}{\text{area}(S > 0)/\epsilon^2} = \log \frac{\text{area}(T > 0)}{\text{area}(S > 0)}, \quad (5)$$

which shows Δy is a constant determined by the vessel area of S and T . This indicates that two fitted lines (respectively for ground truth T and segmentation S) are approximately parallel, i.e. approximate fractal dimensions $f(T)$ and $f(S)$. In this case, the error by $|f(T) - f(S)|$ is tiny even when S is far from T . A network θ trained by minimising $|f(T) - f(S)|$ has gradient $\frac{\partial |f(T) - f(S)|}{\partial \theta} \approx 0$, which faces the issue of gradient vanishing.

Instead, we identify that the box counts $N(T, \epsilon)$ and $N(S, \epsilon)$ are appropriate candidates for loss construction because (1) accurate $N(S, \epsilon)$ derives precise $f(s)$ as $f(s)$ is approximated by the fitting slope between $N(S, \epsilon)$ and ϵ , (2) it avoids the gradient vanishing issue as $|N(T, \epsilon) - N(S, \epsilon)|$ monotonically increases when S is more different from T and (3) the varying box size ϵ introduces regularisation among pixels in various size of receptive fields, i.e. large boxes contribute to semantic information in large receptive fields, while small boxes contain high resolution information in local areas. We calculate the box count loss function $Loss_B(S, T)$ with a set of box sizes $\epsilon = \{2^i | i \in \mathbb{Z}, 2 \leq 2^i \leq \min\{h, w\}\}$ as

$$Loss_B(S, T) = \frac{1}{\sqrt{\sum_i \epsilon_i^2}} \cdot \sqrt{\sum_i \epsilon_i \cdot \left(\frac{N(T, \epsilon_i) - N(S, \epsilon_i)}{N(T, \epsilon_i)} \right)^2}, \quad (6)$$

where $\frac{N(T, \epsilon) - N(S, \epsilon)}{N(T, \epsilon)}$ normalises multi-scale box errors to have the same magnitudes. ϵ_i is empirically configured to weight more on the error of large-size box to regularise semantic information. $1/\sqrt{\sum_i \epsilon_i^2}$ scales $Loss_B(S, T)$ to the same level of magnitude to $Loss_V(S, T)$.

For multi-class vessel segmentation $Loss_B(S_m, T_m)$, we sum up the box count loss functions for artery $Loss_B(S_a, T_a)$ and vein $Loss_B(S_v, T_v)$.

3.3. Differentiable CF-loss for model training

All the calculation for vessel density $v(S)$ and box count $N(S, \epsilon)$ needs to be differentiable to ensure the gradients, as shown in Algorithm 1. We approximate the box count $N(S, \epsilon)$ with two kinds of soft

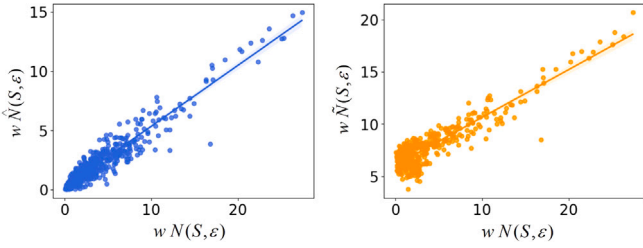


Fig. 3. Linear regression of $Loss_B$ with (w) different box counts. x axis is $Loss_B$ with raw box count $N(S, \epsilon)$ and y axis with the first soft box count $\hat{N}(S, \epsilon)$ (left side) and the second soft box count $\tilde{N}(S, \epsilon)$ (right side).

box count $\hat{N}(S, \epsilon)$ and $\tilde{N}(S, \epsilon)$ as the raw box count $N(S, \epsilon)$ is not differentiable. By combining the operation of *zeropad* and *averagepool*, the vessel ratio in each box can be calculated to select the target boxes. The first soft box count $\hat{N}(S, \epsilon)$ substitutes the binary threshold with an average vessel ratio, while the second soft box count calculates box-wise distance between S and T by $|\hat{N}(S, \epsilon) - \hat{N}(T, \epsilon)|$ and then conducts the summation across all boxes. We randomly sample 1000 pairs of patches (S, T) from multi-class vessel maps and calculate the box count loss $Loss_B(S, T)$ respectively with $N(S, \epsilon)$, $\hat{N}(S, \epsilon)$, and $\tilde{N}(S, \epsilon)$. The correlation is shown in Fig. 3, where the $Loss_B$ with two soft box counts approximate linear functions of $Loss_B$ with the raw box counts.

In model training, the inputs of the proposed loss functions $Loss_V$ and $Loss_B$ are the multi-class vessel segmentation map S_m after *softmax* activation and ground-truth map T_m . $Loss_B$ and $Loss_V$ can be used for end-to-end network training. A combination of pixel-based loss function and complementary regularisation is effective and widely employed in previous work (Shit et al., 2021; Zheng et al., 2021). To simultaneously achieve accurate multi-class segmentation maps and vascular features, we combine the feature-based loss functions, $Loss_V$ and $Loss_B$, and pixel-based cross entropy loss ($Loss_{CE}$) to construct CF-Loss in three ways

$$Loss_{CF-B} = Loss_{CE} + \lambda \cdot Loss_B, \quad (7)$$

$$Loss_{CF-V} = Loss_{CE} + \beta \cdot Loss_V, \quad (8)$$

$$Loss_{CF-VB} = Loss_{CE} + \lambda \cdot Loss_B + \beta \cdot Loss_V, \quad (9)$$

where λ and β are the loss weights for vessel density loss $Loss_V$ and box count loss $Loss_B$ respectively. By adjusting λ and β , the effects of introducing either a single feature-based loss $Loss_V$ and $Loss_B$, or their combination can be investigated.

4. Experiments

4.1. Experiment data

We use three publicly available datasets, DRIVE-AV (Staal et al., 2004; Hu et al., 2013), LES-AV (Orlando et al., 2018), and HRF-AV (Budai et al., 2013; Hemelings et al., 2019), to verify the performance of multi-class vessel segmentation and feature measurement. All three datasets provide the labels for the background, artery, vein, and uncertain classes. Following the labelling strategy of the HRF-AV dataset, all unidentifiable vessels and crossings between arteries and veins are defined as uncertain class. DRIVE-AV has 40 colour fundus photographs with size (565, 584) collected with CR5 non-mydratic 3CCD camera (Canon), where 20 images are for training and 20 for testing. LES-AV contains 22 images with size (1620, 1444), 11 for training and 11 for testing. The imaging device for LES-AV is Visucam Pro NM fundus camera (Carl Zeiss Meditec, Jena, Germany). HRF-AV includes 45 images with size (3504, 2336) imaged with CF-60UVi camera (Canon), 24 for training and 21 for testing. Additionally, the images of DRIVE-AV and

HRF-AV are macula-centred while those of LES-AV are disc-centred, imaging the retinal vessel in different areas. With diverse characteristics of the three datasets, the efficacy of the proposed CF-Loss can be evaluated on test data in different image contrast, size, and laterality.

We include a subset of data from the AlzEye project (Wagner et al., 2022) to verify the clinical contribution brought by CF-Loss. The AlzEye project is a retrospective cohort study of patients aged 40 years and over who have attended Moorfields Eye Hospital between 1 January 2008 and 1 April 2018. Patients were included if they had attended the glaucoma, retina, neuro-ophthalmology or emergency ophthalmic services. Systemic health data were derived from Hospital Episode Statistics data relating to admitted patient care, with a focus on cardiovascular disease and all-cause dementia (Wagner et al., 2022). Specifically, ischaemic stroke was defined as code I63/I64 according to the International Classification of Diseases 10th revision (Schnier et al., 2019). Here, we set a task of predicting three-year ischaemic stroke incidence using derived retinal vascular features. Macula-centred retinal colour images of 1548 patients (774 with stroke and 774 controls) originating from the AlzEye project are used in this task. All the images were from imaging device Topcon 3DOCT-2000SA and collected from the left eye. A logistic regression model was trained with 60% of the data to predict stroke incidence with the input of a single vascular feature, such as fractal dimension or vessel density, and tested with 40% data.

4.2. Backbone network and training details

CF-Loss can be used in any network training. We use U-Net (Ronneberger et al., 2015) and BF-Net (Zhou et al., 2021) as example segmentation backbones. We substitute each backbone's original loss respectively with CF-Loss and compared recent losses, Active Contour (AC) (Chen et al., 2019), GraphCut (GC) (Zheng et al., 2021), cDice (Shit et al., 2021). AC incorporated area and size information of regions of interest in loss function. GC transferred graph cut cost function as an extra penalty to optimise the boundary accuracy. cDice extracted skeleton and regulated network training with Dice score on the skeleton. All compared methods achieved competing performance in a wide range of medical tasks, including retinal binary vessel segmentation. However, the multi-class vessel segmentation in our experiment is much more challenging as the arteries and veins share a highly similar distribution in pixel density and morphology. For each compared loss function, we use their official code and calculate the loss respectively for arteries and veins and sum them up, following the same way of calculating $Loss_V$ and $Loss_B$. As cDice is computationally expensive, we halve BF-Net channels for all loss functions for a fair comparison.

We set the final activation function as softmax. The segmentation output is a class probability map including four categories, background, artery, vein, and uncertain pixel. The uncertain category refers to the pixels that cannot be discriminated due to limited information, such as the intersections between arteries and veins.

In training, all training images are resized to (720, 720) to fit computation resources. For hyperparameters, the learning rate is 0.0008, batch size is 2, and the total epoch is 500. The optimiser is Adam (Kingma and Ba, 2014) with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. We use flipping, rotation, and colour enhancement for data augmentation. 10% training images are used for validation to schedule the learning rate and training process. The learning rate halves when the training loss has stopped decreasing for continuously 50 epochs. When the loss has not decreased for 200 epochs, the training will be stopped. And the model is saved when the validation achieves a higher F1-score (Dice). We report the performance in mean $\pm$ sd of eight models with different subsets of training data. All the performance improvements (%) reported in Section 4.4 are absolute changes. For loss weight λ and β , we respectively set it as 0.5 and 1 based on the highest F1-score in validation data, whilst test data remains unseen in model development.

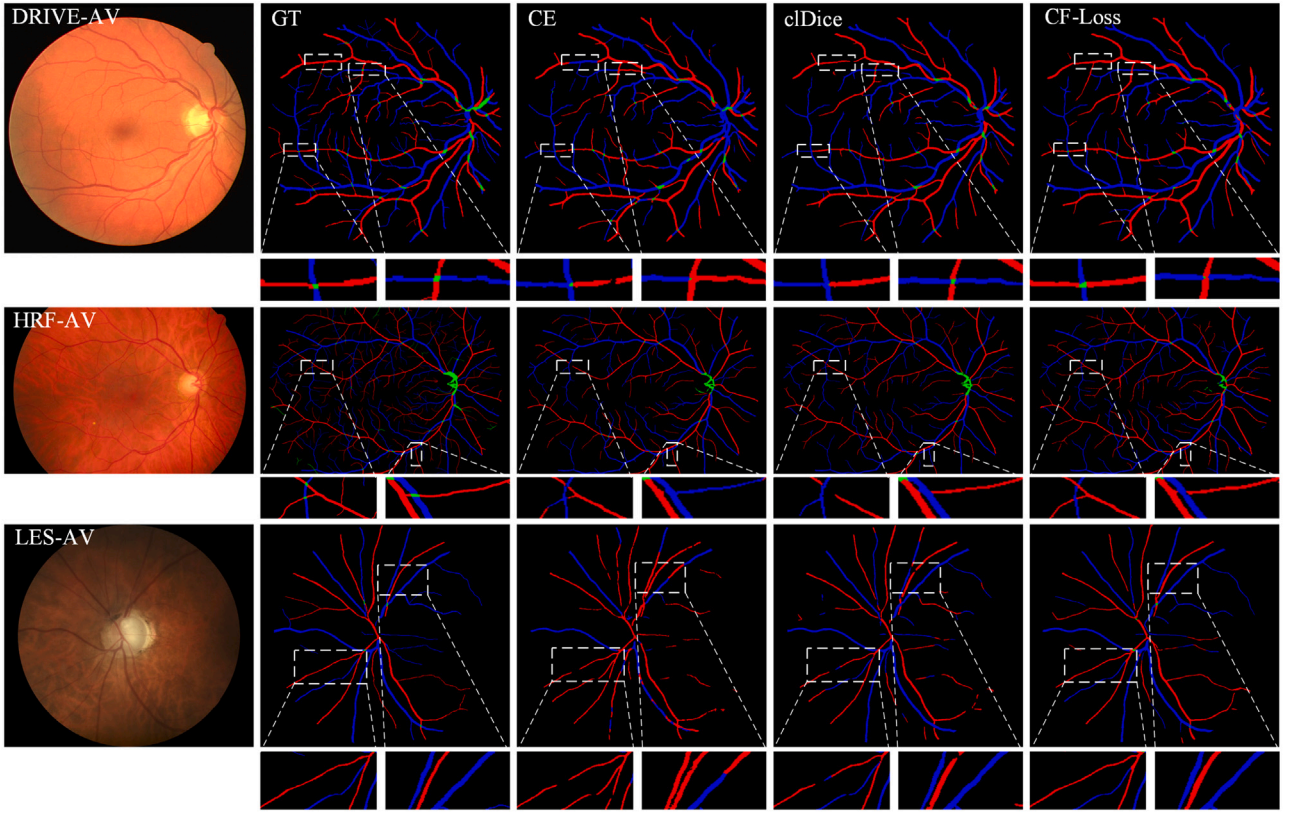


Fig. 4. Visualisation examples on DRIVE-AV, HRF-AV, and LES-AV. From left to right are colour fundus photographs, ground-truth map, segmentation with cross entropy, with cDice, and with a representative of proposed CF-Loss ($Loss_{CF-B}$ in Eq. (7)). Local comparisons are highlighted in the white dash boxes.

The majority of experiment results with $Loss_{CF-B}$ are based on the second soft box count $\tilde{N}(S, \epsilon)$, as it performs better in validation. We implement the code using Pytorch 1.9 and use Tesla T4 GPU (16 GB) in all experiments.

4.3. Evaluation metrics

We evaluate the standard multi-class segmentation performance, instead of artery/vein classification. This can precisely evaluate how the methods perform in segmenting multi-class vessels. The segmentation map is resized back to the original size for calculating metrics. After the maximisation of generated softmax probability map, each pixel has been classified as one of the four categories, the background, artery, vein, and uncertain pixel. We calculate the F1-score (Dice) with $F = (n_a \cdot F_a + n_v \cdot F_v + n_u \cdot F_u) / (n_a + n_v + n_u)$, where n_a , n_v , and n_u respectively represent the pixels number belonging to artery, vein, and uncertainty pixel in label. F_a , F_v , and F_u are the F1-score scores in each binary measurement, e.g. artery pixels versus all other pixels. We also calculate the sensitivity, mean square error (MSE), intersection over union (IOU) for multi-class vessel segmentation.

For evaluating derived vascular features, we measure the agreement of estimated vascular features to ground-truth features derived from manual multi-class annotation. Following previous work (Cheung et al., 2021), intra-class correlation coefficient (ICC) is used to evaluate the feature agreement. Specifically, we evaluate ICC for fractal dimension and vessel density for artery and vein. To study the efficacy of CF-Loss on topology correctness, we measure the Betti number error (Hu et al., 2019; Shit et al., 2021).

To evaluate the real-world clinical benefits by CF-Loss, we show the stroke classification performance of the logistic regression model by AUC-ROC and AUC-PR. p -values from the Mann-Whitney U test are reported, when statistical comparisons are made.

4.4. Experiment results

4.4.1. Comparison among three CF-loss

We first compare the performance of three formats of CF-Loss (Eqs. (7), (8), and (9)). Table 1 and Table 3 show the multi-class vessel segmentation results on three datasets, respectively with backbone U-Net and BF-Net. With U-Net (Table 1), CF-Loss $Loss_{CF-V}$ achieved the highest pixel-based segmentation metrics, including sensitivity, F1-score, IOU, MSE. $Loss_{CF-V}$ also performed well in topology correctness, ranking first place in DRIVE-AV and HRF-AV datasets. For feature agreement of fractal dimension ICC_F and vessel density ICC_V , the CF-Loss $Loss_{CF-VB}$ achieved the best performance on three datasets. The $Loss_{CF-B}$ ranked in the middle for both pixel-based segmentation metrics and feature agreement. When the network backbone is BF-Net (Table 3), the CF-Loss $Loss_{CF-B}$ and $Loss_{CF-VB}$ showed comparable pixel-based segmentation performance, better than $Loss_{CF-V}$ in DRIVE-AV and LES-AV datasets. $Loss_{CF-B}$ also achieved the highest feature agreement across the three datasets. This observation extends to the artery and vein segmentation metrics (Table 4 and Table 5). We select $Loss_{CF-B}$ as a representative for comparison with recent loss functions and the following stroke prediction.

4.4.2. Comparison to recent loss functions

From Table 1 and Table 3, all three CF-Loss showed better performance in all metrics compared to AC, GC, and cDice. With U-Net, $Loss_{CF-B}$ increased F1-score by 2.88%, 2.06%, and 2.03% respectively on DRIVE-AV, LES-AV, and HRF-AV datasets compared to cDice. $Loss_{CF-B}$ highly improved the feature agreement ICC_V by 8%, 7%, and 8% and decreased the Betti error by 3.06, 3.66, and 2.78 respectively on three datasets, compared to cDice. With BF-Net, $Loss_{CF-B}$ increased F1-score by 2.77%, 2.67%, and 2.36% and feature agreement ICC_V by 9%, 12%, and 12% respectively on three datasets, compared to cDice. This improvement in multi-class segmentation performance comprises

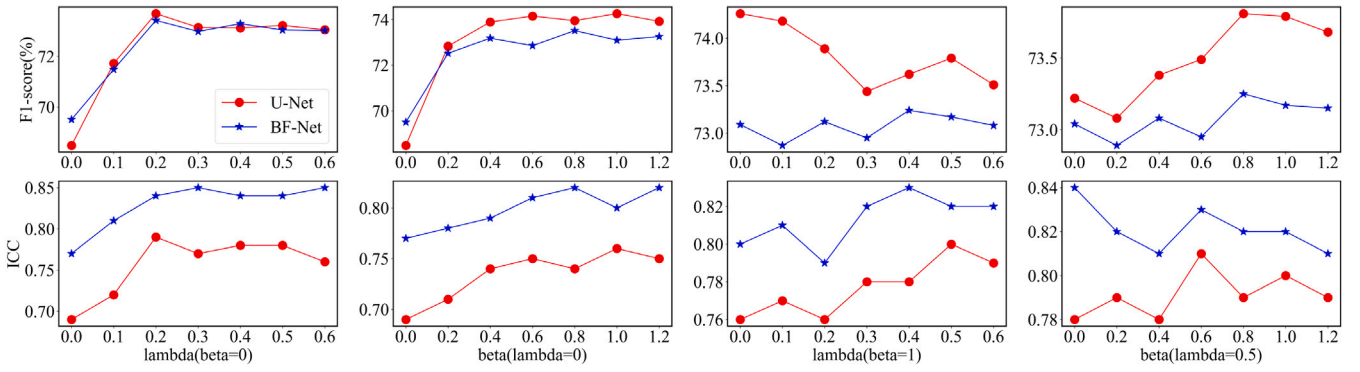


Fig. 5. Performance on DRIVE-AV with different λ and β values for the CF-Loss $Loss_{CF-B}$ in first column, $Loss_{CF-V}$ in second column, and $Loss_{CF-V}$ in last two columns. First row indicates F1-score and second row shows the agreement of fractal dimension ICC_F to ground truth.

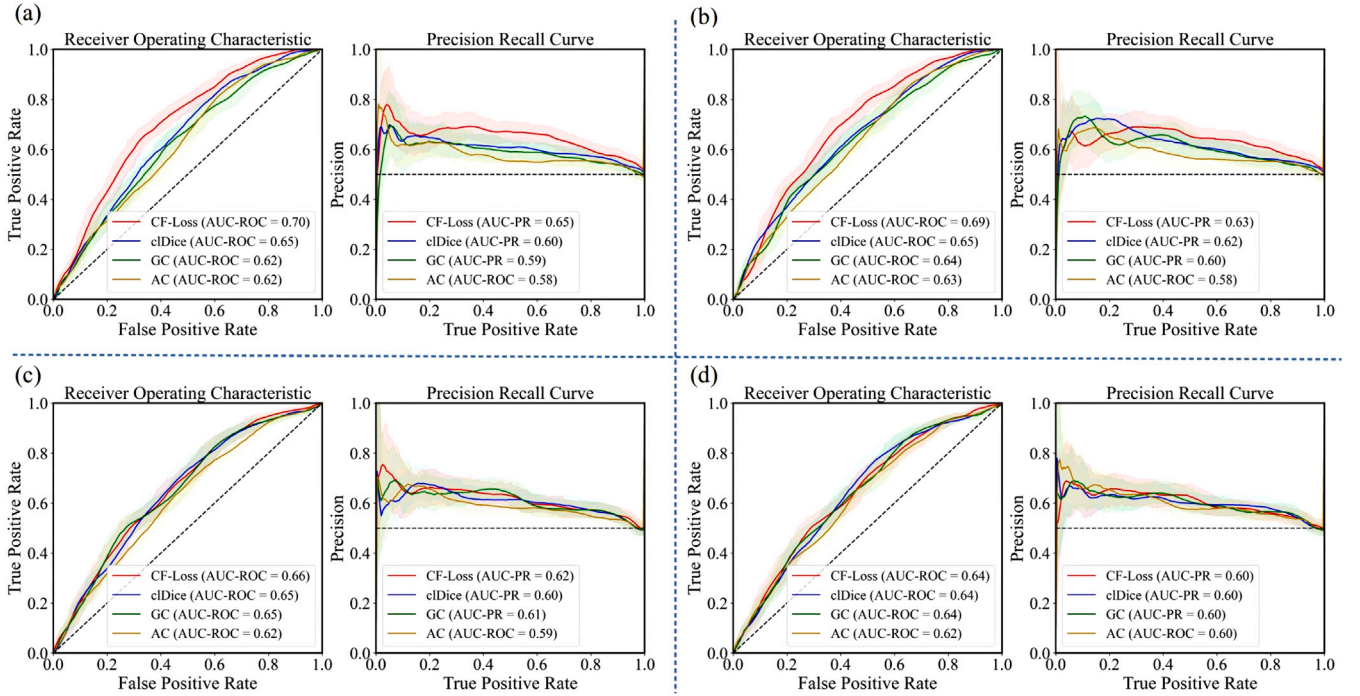


Fig. 6. ROC and PR curves for predicting stroke incidence with (a) artery fractal dimension, (b) artery vessel density, (c) vein fractal dimension, and (d) vein vessel density from different segmentation loss functions. AUC-ROC and AUC-PR are listed in legends. Bootstrapped confidence intervals, [5th, 95th] percentiles of AUC-ROC and AUC-PR, are plotted in corresponding colour shades.

the advances in segmenting arteries and veins (Table 4 and Table 5). We show three segmentation examples for three datasets in Fig. 4. Multi-class vessel maps from CF-Loss $Loss_{CF-B}$ are closest to ground truth maps. The patches highlighted with white dash boxes show that $Loss_{CF-B}$ helps with vessel connectivity and correctness of arteries and veins.

4.4.3. Ablation 1: Impact of loss weights

The loss weights λ and β were set according to the validation performance. We study their effect on test performance. Fig. 5 depicts the performance on DRIVE-AV dataset, with backbone U-Net (red line) and BF-Net (blue line), with different loss weights λ and β . When $\lambda = 0$ and $\beta = 0$, it reduces to cross entropy (CE) loss. The introduction of CF-Loss $Loss_{CF-B}$ ($\lambda = 0.1$ and $\lambda = 0.2$ in first column of Fig. 6) rapidly increases F1-score and fractal dimension agreement. After λ is larger than 0.2, the performance gets saturated. For $Loss_{CF-V}$ in second column, the F1-score and ICC became stable when $\beta = 0.6$.

The performance of $Loss_{CF-V}$ is relatively stable, as shown in the third and fourth columns. With the increase of λ (larger weights

for $Loss_{CF-B}$), ICC_F shows a growing trend while F1-score slightly declines. From the fourth column, β (loss weights for $Loss_{CF-V}$) works in the opposite. This is also supported by Table 1 and Table 3 where $Loss_{CF-B}$ usually performed better in feature agreement while $Loss_{CF-V}$ showed advantages in pixel-based segmentation metrics in some cases. The combination of λ and β presumably offers a weak tradeoff between segmentation metrics and vascular feature agreement on test data.

4.4.4. Ablation 2: Effects of soft box count and box size weight

We compare the performance of two soft box counts, $\hat{N}(S, \epsilon)$ and $\tilde{N}(S, \epsilon)$, which are introduced in Section 3.3. The comparison results on DRIVE-AV dataset is shown in Fig. 7, between the blue boxes ($\tilde{N}(S, \epsilon)$) and green boxes ($\hat{N}(S, \epsilon)$) in each sub-graph. The soft box count $\hat{N}(S, \epsilon)$ provided a higher F1-score and lower MSE than $\tilde{N}(S, \epsilon)$.

In Eq. (6), we empirically use the box size as weights for $Loss_B$. To validate the efficacy of box size weight, we compare the performance of $Loss_{CF-B}$ with- and without box size weight, i.e. $Loss_{B'} = \sqrt{\sum_i \left(\frac{N(T, \epsilon_i) - N(S, \epsilon_i)}{N(T, \epsilon_i)} \right)^2}$. The results are shown in Fig. 7. With the box

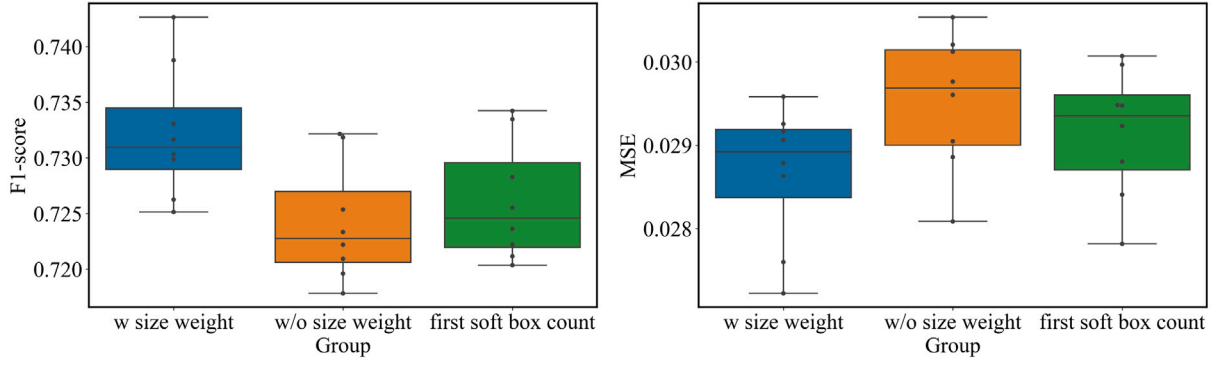


Fig. 7. Performance comparison between $Loss_{CF-B}$ with- and without box size weight, as well as using first soft box count $\hat{N}(S, \epsilon)$ on DRIVE-AV dataset.

Table 1

Multi-class segmentation performance with U-Net on DRIVE-AV, LES-AV, and HRF-AV. Betti error evaluates the topological correctness of segmentation maps. ICC_F evaluate the agreement of fractal dimension to that derived from ground-truth maps and ICC_V for vessel density. p -value of Mann-Whitney U test between CF-Loss $Loss_{CF-B}$ and cDice is reported, as cDice is the most competitive loss function among others.

DRIVE-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	65.13 $\pm$ 1.85	68.4 $\pm$ 1.42	53.29 $\pm$ 1.69	3.18 $\pm$ 0.13	10.64 $\pm$ 0.84	0.58(0.16–0.77)	0.55(0.12–0.75)
GC (Zheng et al., 2021)	65.94 $\pm$ 1.08	69.23 $\pm$ 2.05	53.03 $\pm$ 2.37	3.27 $\pm$ 0.22	23.22 $\pm$ 2.34	0.64(0.27–0.81)	0.62(0.25–0.82)
cDice (Shit et al., 2021)	68.78 $\pm$ 0.96	70.34 $\pm$ 1.11	55.64 $\pm$ 1.38	3.05 $\pm$ 0.15	10.98 $\pm$ 0.92	0.61(0.32–0.78)	0.64(0.28–0.83)
CF-Loss $Loss_{CF-B}$	71.37 $\pm$ 0.75	73.22 $\pm$ 0.98	58.25 $\pm$ 1.21	2.85 $\pm$ 0.01	7.92 $\pm$ 1.02	0.78(0.45–0.92)	0.72(0.36–0.93)
CF-Loss $Loss_{CF-V}$	73.27 $\pm$ 1.24	74.26 $\pm$ 0.66	59.5 $\pm$ 0.84	2.77 $\pm$ 0.06	6.73 $\pm$ 0.46	0.76(0.42–0.92)	0.69(0.38–0.86)
CF-Loss $Loss_{CF-VB}$	71.93 $\pm$ 0.11	73.79 $\pm$ 0.81	58.93 $\pm$ 1.02	2.78 $\pm$ 0.11	7.07 $\pm$ 1.32	0.83(0.55–0.91)	0.86(0.62–0.92)
p -value	2.39e–2	1.36e–3	5.38e–3	2.39e–2	1.36e–3	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	59.15 $\pm$ 2.57	62.83 $\pm$ 2.32	47.4 $\pm$ 2.6	2.88 $\pm$ 0.16	8.42 $\pm$ 0.75	0.67(0.24–0.84)	0.62(0.21–0.82)
GC (Zheng et al., 2021)	60.32 $\pm$ 1.98	63.69 $\pm$ 1.78	47.99 $\pm$ 1.91	2.83 $\pm$ 0.14	10.69 $\pm$ 2.84	0.67(0.18–0.86)	0.62(0.22–0.8)
cDice (Shit et al., 2021)	61.04 $\pm$ 2.49	63.87 $\pm$ 1.94	48.55 $\pm$ 1.9	2.86 $\pm$ 0.12	8.42 $\pm$ 1.31	0.66(0.28–0.81)	0.64(0.35–0.77)
CF-Loss $Loss_{CF-B}$	62.21 $\pm$ 2.14	65.93 $\pm$ 1.32	50.66 $\pm$ 1.51	2.61 $\pm$ 0.25	4.76 $\pm$ 1.15	0.72(0.33–0.94)	0.71(0.32–0.9)
CF-Loss $Loss_{CF-V}$	62.47 $\pm$ 2.52	66.4 $\pm$ 1.74	51.11 $\pm$ 2.64	2.56 $\pm$ 0.16	5.34 $\pm$ 1.15	0.69(0.22–0.85)	0.7(0.38–0.89)
CF-Loss $Loss_{CF-VB}$	61.51 $\pm$ 1.56	64.71 $\pm$ 1.44	49.7 $\pm$ 1.89	2.69 $\pm$ 0.18	4.78 $\pm$ 1.07	0.78(0.47–0.91)	0.73(0.42–0.9)
p -value	1.36e–3	2.17e–3	7.29e–3	2.15e–2	3.13e–3	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	65.37 $\pm$ 1.36	69.93 $\pm$ 0.98	55.11 $\pm$ 0.88	2.11 $\pm$ 0.05	9.31 $\pm$ 0.48	0.78(0.59–0.85)	0.72(0.39–0.85)
GC (Zheng et al., 2021)	66.75 $\pm$ 1.21	70.48 $\pm$ 0.63	55.73 $\pm$ 0.82	2.18 $\pm$ 0.07	12.84 $\pm$ 2.4	0.74(0.51–0.83)	0.69(0.36–0.84)
cDice (Shit et al., 2021)	67.55 $\pm$ 1.92	70.14 $\pm$ 1.04	56.12 $\pm$ 0.67	2.13 $\pm$ 0.04	9.39 $\pm$ 0.46	0.78(0.62–0.83)	0.73(0.63–0.92)
CF-Loss $Loss_{CF-B}$	69.41 $\pm$ 1.75	72.17 $\pm$ 0.66	57.74 $\pm$ 0.73	1.91 $\pm$ 0.02	6.61 $\pm$ 0.52	0.82(0.68–0.92)	0.81(0.65–0.91)
CF-Loss $Loss_{CF-V}$	69.85 $\pm$ 1.7	72.47 $\pm$ 0.38	58.06 $\pm$ 0.41	1.89 $\pm$ 0.05	6.38 $\pm$ 0.39	0.78(0.65–0.88)	0.79(0.57–0.95)
CF-Loss $Loss_{CF-VB}$	68.89 $\pm$ 2.24	71.82 $\pm$ 1.14	57.3 $\pm$ 0.94	1.93 $\pm$ 0.05	6.49 $\pm$ 0.69	0.84(0.71–0.95)	0.83(0.64–0.94)
p -value	2.17e–3	1.36e–3	1.36e–3	9.39e–4	3.67e–3	N/A	N/A

Table 2

Effects of single loss function on multi-class vessel segmentation performance with U-Net on DRIVE-AV dataset. $Loss_B$ indicates box count loss only and $Loss_{CF-B}$ is with the combination of cross entropy and $Loss_B$, as introduced in Eq. (7).

Loss	F1-score $\uparrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$
$Loss_{CE}$	69.14 $\pm$ 1.57	18.22 $\pm$ 2.63	0.58(0.25–0.72)
$Loss_B$	61.31 $\pm$ 0.71	16.69 $\pm$ 1.19	0.52(0.14–0.75)
$Loss_{CF-B}$	73.22 $\pm$ 0.98	7.92 $\pm$ 1.02	0.78(0.45–0.92)

size weight, the performance has been moderately improved, which validated that emphasised semantic information helps multi-class vessel segmentation.

4.4.5. Ablation 3: Effects of single loss function

In Section 4.4.3, we have verified that feature-based loss $Loss_V$ and $Loss_B$ help the model performance. We here study the performance of a single feature-based loss function to understand the benefits of combining a pixel-based loss. Table 2 shows the distribution of performance achieved by $Loss_B$ and $Loss_{CF-B}$. The $Loss_B$ alone works poorly

in multi-class vessel segmentation, which highlights that pixel-based segmentation loss, such as CE, is indispensable for segmentation tasks. Considering the mutual benefits, the pixel-based CE and feature-based loss functions appear to be complementary to each other.

4.4.6. Segmentation metrics vs. feature agreement

We identify that high segmentation metrics do not necessarily correspond to accurate vascular features, which breaks the common assumption that purely optimising pixel-based metrics necessarily provide accurate features. We found three aspects supporting our observation. The first aspect is the network backbone. Comparing Table 1 and Table 3, U-Net achieved higher pixel-based segmentation metrics, such as F1-score and IOU, while BF-Net offered higher feature agreement, ICC_F and ICC_V on DRIVE-AV dataset. The relationship is also visualised in Fig. 5. The second aspect is segmentation categories. According to Table 4 and Table 5, arteries showed lower segmentation metrics but higher ICC_F and ICC_V than veins in most of the cases. For example, $Loss_{CF-B}$ achieved 5.14% higher F1-score in veins but 13% higher ICC_F in arteries on DRIVE-AV dataset. The final aspect is loss function. In Table 2, $Loss_{CF-V}$ showed better segmentation metrics but lower

Table 3

Multi-class segmentation performance with BF-Net on DRIVE-AV, LES-AV, and HRF-AV. Betti error evaluates the topological correctness of segmentation maps. ICC_F evaluate the agreement of fractal dimension to that derived from ground-truth maps and ICC_V for vessel density. p -value of Mann-Whitney U test between CF-Loss $Loss_{CF-B}$ and cDice is reported, as cDice is the most competitive loss function among others.

DRIVE-AV							
Loss	Sensitivity $\uparrow$	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	64.15 $\pm$ 2.78	67.31 $\pm$ 2.08	52.22 $\pm$ 1.97	3.31 $\pm$ 0.32	14.04 $\pm$ 3.91	0.62(0.31–0.76)	0.61(0.23–0.81)
GC (Zheng et al., 2021)	66.93 $\pm$ 2.42	68.87 $\pm$ 2.48	52.57 $\pm$ 2.63	3.29 $\pm$ 0.35	13.96 $\pm$ 4.69	0.66(0.26–0.82)	0.68(0.35–0.76)
cDice (Shit et al., 2021)	68.31 $\pm$ 1.88	70.27 $\pm$ 1.1	54.55 $\pm$ 1.36	3.22 $\pm$ 0.18	11.48 $\pm$ 1.54	0.71(0.28–0.88)	0.69(0.26–0.87)
CF-Loss $Loss_{CF-B}$	72.91 $\pm$ 1.27	73.04 $\pm$ 0.58	57.99 $\pm$ 0.7	2.93 $\pm$ 0.06	7.75 $\pm$ 1.21	0.84(0.64–0.92)	0.78(0.56–0.92)
CF-Loss $Loss_{CF-V}$	71.93 $\pm$ 1.16	73.09 $\pm$ 0.55	58.14 $\pm$ 0.65	2.88 $\pm$ 0.09	7.91 $\pm$ 1.34	0.8(0.54–0.94)	0.74(0.51–0.88)
CF-Loss $Loss_{CF-VB}$	72.29 $\pm$ 2.03	73.17 $\pm$ 1.14	58.18 $\pm$ 1.38	2.89 $\pm$ 0.12	7.07 $\pm$ 0.93	0.82(0.54–0.9)	0.76(0.48–0.89)
p -value	1.36e–3	1.36e–3	1.36e–3	3.87e–3	5.07e–3	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_F(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	63.66 $\pm$ 2.17	66.15 $\pm$ 2.45	51.09 $\pm$ 2.8	2.63 $\pm$ 0.27	9.3 $\pm$ 2.86	0.69(0.33–0.94)	0.72(0.26–0.94)
GC (Zheng et al., 2021)	62.85 $\pm$ 2.08	65.88 $\pm$ 2.58	50.18 $\pm$ 2.82	2.68 $\pm$ 0.18	6.76 $\pm$ 2.58	0.83(0.61–0.92)	0.83(0.57–0.94)
cDice (Shit et al., 2021)	64.16 $\pm$ 1.99	67.2 $\pm$ 2.23	51.87 $\pm$ 2.3	2.5 $\pm$ 0.19	4.51 $\pm$ 0.69	0.84(0.58–0.94)	0.83(0.61–0.92)
CF-Loss $Loss_{CF-B}$	67.06 $\pm$ 1.76	69.87 $\pm$ 1.56	54.98 $\pm$ 1.61	2.32 $\pm$ 0.1	3.04 $\pm$ 0.66	0.92(0.83–0.97)	0.95(0.88–0.98)
CF-Loss $Loss_{CF-V}$	65.85 $\pm$ 1.81	69.05 $\pm$ 1.45	54.15 $\pm$ 1.59	2.35 $\pm$ 0.13	3.12 $\pm$ 0.71	0.91(0.8–0.97)	0.93(0.83–0.93)
CF-Loss $Loss_{CF-VB}$	66.83 $\pm$ 1.24	69.89 $\pm$ 0.9	54.95 $\pm$ 0.87	2.31 $\pm$ 0.06	2.71 $\pm$ 0.59	0.88(0.72–0.95)	0.9(0.78–0.96)
p -value	1.36e–3	1.95e–3	9.39e–4	5.38e–3	3.06e–3	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	$ICC_A(95\%CI) \uparrow$	$ICC_V(95\%CI) \uparrow$
AC (Chen et al., 2019)	65.19 $\pm$ 2.79	68.22 $\pm$ 0.8	53.09 $\pm$ 0.73	2.26 $\pm$ 0.04	10.81 $\pm$ 3.59	0.78(0.43–0.92)	0.81(0.42–0.94)
GC (Zheng et al., 2021)	65.74 $\pm$ 2.51	68.67 $\pm$ 1.06	53.77 $\pm$ 1.28	2.15 $\pm$ 0.05	12.93 $\pm$ 4.32	0.76(0.41–0.91)	0.77(0.32–0.92)
cDice (Shit et al., 2021)	65.66 $\pm$ 2.18	68.83 $\pm$ 0.45	54.04 $\pm$ 0.57	2.13 $\pm$ 0.05	9.48 $\pm$ 1.1	0.81(0.38–0.96)	0.79(0.46–0.92)
CF-Loss $Loss_{CF-B}$	67.61 $\pm$ 2.48	71.19 $\pm$ 0.58	56.48 $\pm$ 0.73	1.96 $\pm$ 0.03	6.82 $\pm$ 1.16	0.86(0.71–0.96)	0.91(0.82–0.96)
CF-Loss $Loss_{CF-V}$	71.14 $\pm$ 1.87	72.31 $\pm$ 0.71	57.76 $\pm$ 0.91	1.96 $\pm$ 0.05	7.68 $\pm$ 0.9	0.82(0.68–0.91)	0.86(0.72–0.93)
CF-Loss $Loss_{CF-VB}$	68.29 $\pm$ 2.34	71.39 $\pm$ 0.77	56.74 $\pm$ 0.94	1.95 $\pm$ 0.04	7.11 $\pm$ 1.36	0.83(0.71–0.92)	0.85(0.76–0.93)
p -value	5.38e–3	9.39e–4	9.39e–4	9.39e–4	2.76e–3	N/A	N/A

ICC_F and ICC_V than $Loss_{CF-VB}$ on HRF-AV dataset. When the loss weight λ and β change for $Loss_{CF-VB}$, F1-score and ICC_F show contrary general trend, depicted in the third and fourth columns of Fig. 5.

These three aspects reveal that it is questionable to believe better segmentation metrics correspond to more accurate vascular features. Similarly, previous work found that topology correctness cannot be ensured by pixel-based segmentation metrics and proposed specific regularisation on topology. Here we further discovered the potential inconsistency of the pixel-based segmentation metrics and vascular feature agreement, thus highlighting the importance of directly regulating the features to improve feature accuracy for downstream clinical tasks.

4.4.7. Quantitative clinical impact

We use measured vascular features to predict three-year ischaemic stroke incidence. From Table 1 and Table 3, we can identify that BF-Net performed best in feature agreement ICC_F and ICC_V for all loss functions, on three datasets. Hence, we used the trained BF-Net to segment the ischaemic stroke database and then measure the vascular features, including fractal dimension and vessel density. We train and validate a logistic regression with the input of a single vascular feature and output of stroke incidence.

The results with the input of the fractal dimension and vessel density are shown in Fig. 6. The best prediction performance is achieved with the artery fractal dimension. AUC-ROC achieved 0.7 and AUC-PR achieved 0.65 with CF-Loss $Loss_{CF-B}$, outperforming those of the other three compared loss functions, depicted in Fig. 6(a). This demonstrates that CF-Loss quantitatively improved downstream task performance with accurate vascular features. The logistic regression model with input of artery vessel density also achieved AUC-ROC of 0.69 (Fig. 6(b)), clearly outperforming the models based on other loss functions. For the vein fractal dimension and vessel density, we observe that ROC and PR curves from the four loss functions largely overlap, as shown in Fig. 6(c) and 6(d), despite the clear difference in veins' features in Table 4 and Table 5. This suggests that ischaemic stroke prediction is not sensitive to veins' fractal dimension and vessel density. Considering that arteries' features contribute to better prediction performance,

arteries are likely to be better markers for ischaemic stroke prediction in the collected database. This is consistent with the findings of recent work (Sandoval-Garcia et al., 2021).

4.4.8. Computation efficiency

MI-Loss requires $O(n^2)$ computational complexity for a two-dimensional image with a size of $n \times n$, which is at the same level of complexity as AC, GC, and cDice. For model training, the proposed MI-Loss $Loss_{MI-B}$ on average took 1.64 s for a batch size of 2 and image size of (720, 720), while AC, GC, and cDice respectively took 1.87 s, 1.91 s, and 2.17 s. The inference time for MI-Loss and compared loss functions is equal.

5. Discussion and conclusion

We introduce a new feature-based loss function that directly encodes computation of downstream vascular features used as clinical disease markers, to optimise segmentation networks. We combine the proposed feature-based loss and pixel-based loss as CF-Loss. Experimental results show that CF-Loss improved both multi-class vessel segmentation and vascular feature measurement, compared to three recent loss functions which consider only the interim segmentation rather than the final feature. In particular, the combination of $Loss_{CF-B}$ and BF-Net achieved the best results in feature measurement. With CF-Loss, both artery and vein segmentation have been improved, as shown in Table 4 and Table 5. We also conduct three ablation study to show the efficacy of designed components, including single loss function and box size weight. The analysis of segmentation metrics vs. feature agreement (Section 4.4.6) highlights the significance of optimising features directly, with CF-Loss proposed in this work. For the loss weight λ and β respectively for $Loss_{CF-B}$ and $Loss_{CF-V}$, there appears to be a trade-off of further optimising pixel-based segmentation metrics and vascular feature agreement. In scenarios that require accurate features for clinical downstream tasks, such as the ischaemic stroke prediction, the $Loss_{CF-B}$ is preferred.

Future work will focus on investigating the combination strategy of multiple feature-based loss functions, as well as the principle behind

Table 4

Artery and vein segmentation performance with U-Net on DRIVE-AV, LES-AV, and HRF-AV.

Evaluation category: artery							
DRIVE-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	64.42 ± 1.55	66.91 ± 2.09	51.37 ± 1.41	3.08 ± 0.85	8.84 ± 1.16	0.74(0.36–0.9)	0.73(0.35–0.88)
GC (Zheng et al., 2021)	64.73 ± 1.98	67.35 ± 2.39	52.29 ± 1.46	3.15 ± 0.62	17.21 ± 2.18	0.77(0.44–0.91)	0.75(0.38–0.88)
clDice (Shit et al., 2021)	65.63 ± 2.36	68.37 ± 1.76	53.39 ± 1.89	2.97 ± 0.29	6.9 ± 1.56	0.75(0.36–0.89)	0.77(0.54–0.92)
CF-Loss $Loss_{CF-B}$	67.71 ± 2.47	71.44 ± 1.27	55.78 ± 1.52	2.77 ± 0.16	4.91 ± 1.67	0.8(0.52–0.92)	0.88(0.71–0.95)
CF-Loss $Loss_{CF-V}$	70.56 ± 1.35	72.74 ± 0.84	57.33 ± 1.04	2.71 ± 0.07	4.23 ± 0.65	0.78(0.49–0.93)	0.85(0.68–0.92)
CF-Loss $Loss_{CF-VB}$	69.38 ± 1.41	72.21 ± 1.06	56.69 ± 1.28	2.73 ± 0.15	4.84 ± 2.52	0.87(0.68–0.95)	0.9(0.76–0.96)
p-value	1.81e–2	1.36e–3	2.39e–3	9.39e–4	1.36e–3	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	58.32 ± 2.66	61.83 ± 1.63	46.64 ± 1.83	2.75 ± 0.58	7.38 ± 2.56	0.66(0.31–0.82)	0.58(0.15–0.75)
GC (Zheng et al., 2021)	58.45 ± 2.59	61.83 ± 1.79	46.99 ± 1.29	2.78 ± 0.35	7.84 ± 2.37	0.67(0.12–0.89)	0.58(0.2–0.76)
clDice (Shit et al., 2021)	59.15 ± 1.28	62.14 ± 1.22	47.21 ± 2.04	2.72 ± 0.11	7.53 ± 1.68	0.65(0.24–0.83)	0.59(0.21–0.78)
CF-Loss $Loss_{CF-B}$	61.2 ± 2.63	64.06 ± 1.78	48.25 ± 1.94	2.52 ± 0.08	3.87 ± 1.24	0.72(0.4–0.92)	0.68(0.33–0.84)
CF-Loss $Loss_{CF-V}$	62.49 ± 2.28	64.85 ± 1.38	49.04 ± 1.38	2.44 ± 0.12	3.95 ± 1.07	0.67(0.36–0.84)	0.65(0.33–0.85)
CF-Loss $Loss_{CF-VB}$	61.04 ± 1.68	62.92 ± 1.34	47.1 ± 1.03	2.61 ± 0.14	3.75 ± 0.55	0.77(0.36–0.94)	0.71(0.32–0.92)
p-value	1.81e–2	1.36e–2	2.39e–2	3.88e–3	2.76e–3	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	64.87 ± 2.55	69.61 ± 1.25	54.78 ± 1.39	2.23 ± 0.12	8.17 ± 0.67	0.87(0.69–0.95)	0.78(0.46–0.92)
GC (Zheng et al., 2021)	65.79 ± 2.21	70.25 ± 1.13	55.29 ± 1.48	2.26 ± 0.09	10.67 ± 1.98	0.88(0.8–0.95)	0.73(0.36–0.9)
clDice (Shit et al., 2021)	66.93 ± 1.78	70.38 ± 1.29	55.52 ± 0.88	2.24 ± 0.08	8.24 ± 0.92	0.88(0.82–0.95)	0.79(0.46–0.94)
CF-Loss $Loss_{CF-B}$	69.32 ± 1.95	72.61 ± 0.66	57.2 ± 0.79	2.03 ± 0.04	5.6 ± 0.53	0.91(0.79–0.97)	0.86(0.67–0.95)
CF-Loss $Loss_{CF-V}$	69.54 ± 2.03	72.84 ± 0.42	57.49 ± 0.51	2.01 ± 0.05	5.11 ± 0.85	0.81(0.47–0.85)	0.84(0.79–0.96)
CF-Loss $Loss_{CF-VB}$	68.83 ± 2.22	72.24 ± 1.14	56.76 ± 1.5	2.05 ± 0.06	5.55 ± 0.45	0.92(0.81–0.97)	0.93(0.84–0.97)
p-value	3.88e–3	1.01e–2	2.39e–2	9.39e–4	3.88e–3	N/A	N/A
Evaluation category: vein							
DRIVE-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	71.25 ± 1.66	71.01 ± 1.61	56.47 ± 1.85	3.37 ± 0.12	13.43 ± 0.89	0.52(0.12–0.82)	0.45(0.08–0.76)
GC (Zheng et al., 2021)	71.82 ± 1.81	72.43 ± 1.7	57.51 ± 1.65	3.49 ± 0.09	25.69 ± 2.27	0.58(0.28–0.86)	0.46(0.12–0.75)
clDice (Shit et al., 2021)	73.24 ± 1.18	73.71 ± 1.53	59.47 ± 1.65	3.31 ± 0.11	13.37 ± 1.42	0.55(0.28–0.82)	0.48(0.21–0.88)
CF-Loss $Loss_{CF-B}$	77.18 ± 0.65	76.58 ± 0.76	62.12 ± 1.07	3.18 ± 0.09	11.4 ± 0.89	0.67(0.36–0.87)	0.54(0.28–0.82)
CF-Loss $Loss_{CF-V}$	78.04 ± 1.61	77.31 ± 0.56	63.13 ± 0.74	2.98 ± 0.06	9.27 ± 0.82	0.66(0.15–0.87)	0.56(0.32–0.82)
CF-Loss $Loss_{CF-VB}$	76.45 ± 0.48	76.92 ± 1.01	62.61 ± 1.32	2.98 ± 0.11	9.43 ± 1.71	0.69(0.41–0.88)	0.58(0.33–0.85)
p-value	9.39e–4	9.39e–4	5.38e–3	1.95e–3	7.41e–3	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	60.29 ± 2.72	63.79 ± 2.53	49.47 ± 2.49	3.15 ± 0.18	13.11 ± 1.28	0.67(0.31–0.82)	0.65(0.23–0.79)
GC (Zheng et al., 2021)	61.77 ± 2.35	64.38 ± 1.94	50.75 ± 1.77	3.08 ± 0.29	26.29 ± 2.72	0.68(0.12–0.89)	0.66(0.24–0.83)
clDice (Shit et al., 2021)	61.86 ± 2.48	65.67 ± 1.48	50.7 ± 1.63	3.14 ± 0.13	14.18 ± 1.34	0.67(0.24–0.83)	0.68(0.32–0.81)
CF-Loss $Loss_{CF-B}$	63.39 ± 1.89	68.39 ± 1.37	53.37 ± 1.61	2.71 ± 0.23	5.27 ± 1.68	0.73(0.34–0.91)	0.74(0.28–0.94)
CF-Loss $Loss_{CF-V}$	63.59 ± 2.52	68.57 ± 2.78	53.56 ± 2.64	2.69 ± 0.19	6.5 ± 1.21	0.7(0.25–0.89)	0.75(0.21–0.88)
CF-Loss $Loss_{CF-VB}$	61.84 ± 1.23	66.21 ± 1.75	51.99 ± 1.59	2.94 ± 0.12	5.67 ± 1.64	0.78(0.24–0.94)	0.75(0.16–0.91)
p-value	2.39e–2	7.41e–3	1.01e–2	1.36e–3	9.39e–4	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score ↑	IOU ↑	MSE ↓	Betti error↓	ICC _F (95%CI) ↑	ICC _V (95%CI) ↑
AC (Chen et al., 2019)	71.93 ± 0.98	74.93 ± 0.98	60.±0.88	2.0 ± 0.05	11.31 ± 0.48	0.62(0.22–0.78)	0.61(0.18–0.82)
GC (Zheng et al., 2021)	72.93 ± 0.98	75.48 ± 0.63	61.73 ± 0.82	2.2 ± 0.07	14.84 ± 2.4	0.58(0.24–0.74)	0.62(0.26–0.76)
clDice (Shit et al., 2021)	73.93 ± 0.98	75.14 ± 1.04	61.12 ± 0.67	2.27 ± 0.04	11.39 ± 0.46	0.64(0.21–0.82)	0.66(0.26–0.81)
CF-Loss $Loss_{CF-B}$	76.23 ± 1.93	77.72 ± 0.47	63.67 ± 0.62	1.96 ± 0.02	8.34 ± 0.96	0.71(0.31–0.89)	0.7(0.35–0.88)
CF-Loss $Loss_{CF-V}$	76.73 ± 1.69	77.89 ± 0.36	63.92 ± 0.45	1.95 ± 0.05	8.24 ± 0.33	0.68(0.29–0.88)	0.76(0.42–0.9)
CF-Loss $Loss_{CF-VB}$	75.65 ± 2.45	77.37 ± 0.96	63.23 ± 1.25	1.98 ± 0.06	8.15 ± 0.67	0.73(0.33–0.89)	0.78(0.46–0.91)
p-value	9.39e–4	1.36e–3	9.39e–4	9.39e–4	1.36e–3	N/A	N/A

considering that simple summation of $Loss_B$ and $Loss_V$ in Eq. (9) did not provide clear extra benefit. This raises a few investigation interests. First, a more sophisticated integration might be required to maximise the effects of each component, considering the linear combination cannot further help the performance. The compatibility of the specific format of CF-Loss and network architecture is also worthy of exploration. Although Tables 1 and 3 show that all combinations of CF-Loss and architectures produce better performance than the compared loss functions, the box count loss $Loss_B$ particularly cooperates well with the BF-Net, an information fusion architecture. One reason might be

that multi-size view in box counts of $Loss_B$ enhances the network's ability to learn multi-scale features. Beyond our specific application to two vascular markers from retinal fundus photographs, we will explore encoding additional features into this clinically-relevant feature optimised loss function. Although we show improvements in the accuracy of some features such as vessel density and fractal dimension, some tortuosity and calibre metrics might offer extra regularisation to model training. This will also help explore the best disease markers among a wide range of vascular features. In stroke incidence prediction, we have not adjusted logistic regression model to demographic factors as our target is to compare the derived vascular features from CF-Loss and recent loss

Table 5

Artery and vein segmentation performance with BF-Net on DRIVE-AV, LES-AV, and HRF-AV.

Evaluation category: artery							
DRIVE-AV							
Loss	Sensitivity $\uparrow$	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _F (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	61.85 $\pm$ 2.19	65.16 $\pm$ 2.12	50.15 $\pm$ 1.62	3.21 $\pm$ 0.26	10.27 $\pm$ 2.58	0.64(0.09–0.86)	0.68(0.32–0.84)
GC (Zheng et al., 2021)	64.04 $\pm$ 2.77	66.71 $\pm$ 2.82	50.09 $\pm$ 2.38	3.16 $\pm$ 0.13	10.42 $\pm$ 2.75	0.71(0.27–0.89)	0.74(0.45–0.91)
clDice (Shit et al., 2021)	64.63 $\pm$ 2.54	68.39 $\pm$ 1.62	52.43 $\pm$ 1.47	3.12 $\pm$ 0.14	8.57 $\pm$ 1.18	0.74(0.45–0.9)	0.77(0.49–0.92)
CF-Loss $Loss_{CF-B}$	68.45 $\pm$ 2.47	71.11 $\pm$ 0.93	55.35 $\pm$ 1.12	2.85 $\pm$ 0.09	4.5 $\pm$ 1.25	0.85(0.62–0.94)	0.82(0.57–0.93)
CF-Loss $Loss_{CF-V}$	68.75 $\pm$ 1.98	71.86 $\pm$ 0.51	56.25 $\pm$ 0.63	2.76 $\pm$ 0.07	4.7 $\pm$ 1.06	0.81(0.52–0.93)	0.79(0.49–0.95)
CF-Loss $Loss_{CF-VB}$	69.48 $\pm$ 1.64	71.33 $\pm$ 1.34	55.62 $\pm$ 1.58	2.86 $\pm$ 0.11	4.8 $\pm$ 0.55	0.84(0.62–0.93)	0.82(0.59–0.91)
p-value	9.39e–4	9.39e–4	7.41e–3	1.36e–3	9.39e–4	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _F (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	64.07 $\pm$ 2.16	65.49 $\pm$ 2.28	50.31 $\pm$ 1.92	2.57 $\pm$ 0.18	9.41 $\pm$ 2.67	0.68(0.3–0.92)	0.68(0.28–0.79)
GC (Zheng et al., 2021)	63.24 $\pm$ 2.09	64.18 $\pm$ 1.58	49.53 $\pm$ 1.28	2.53 $\pm$ 0.22	7.34 $\pm$ 2.18	0.83(0.56–0.95)	0.81(0.54–0.92)
clDice (Shit et al., 2021)	65.39 $\pm$ 2.08	65.57 $\pm$ 2.48	50.48 $\pm$ 1.63	2.46 $\pm$ 0.09	5.54 $\pm$ 0.36	0.82(0.47–0.96)	0.82(0.64–0.91)
CF-Loss $Loss_{CF-B}$	68.11 $\pm$ 1.96	68.43 $\pm$ 1.34	52.94 $\pm$ 1.4	2.28 $\pm$ 0.1	3.48 $\pm$ 0.83	0.86(0.51–0.96)	0.91(0.67–0.98)
CF-Loss $Loss_{CF-V}$	66.69 $\pm$ 2.62	67.53 $\pm$ 1.45	51.91 $\pm$ 1.69	2.33 $\pm$ 0.19	3.85 $\pm$ 1.62	0.85(0.52–0.93)	0.89(0.64–0.95)
CF-Loss $Loss_{CF-VB}$	66.4 $\pm$ 1.89	68.17 $\pm$ 0.99	52.52 $\pm$ 1.03	2.26 $\pm$ 0.07	2.81 $\pm$ 0.8	0.85(0.46–0.96)	0.9(0.63–0.97)
p-value	1.95e–3	3.88e–3	3.13e–2	1.36e–2	9.39e–4	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _A (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	65.45 $\pm$ 2.19	67.43 $\pm$ 1.51	52.17 $\pm$ 2.05	2.33 $\pm$ 0.09	7.76 $\pm$ 3.15	0.86(0.68–0.95)	0.87(0.72–0.94)
GC (Zheng et al., 2021)	65.49 $\pm$ 2.66	68.25 $\pm$ 1.65	53.48 $\pm$ 1.75	2.27 $\pm$ 0.12	9.32 $\pm$ 2.36	0.83(0.59–0.93)	0.82(0.61–0.94)
clDice (Shit et al., 2021)	65.53 $\pm$ 1.48	68.15 $\pm$ 1.83	53.23 $\pm$ 1.22	2.29 $\pm$ 0.08	7.43 $\pm$ 1.04	0.89(0.74–0.96)	0.86(0.7–0.96)
CF-Loss $Loss_{CF-B}$	67.83 $\pm$ 2.53	71.42 $\pm$ 1.03	55.75 $\pm$ 1.24	2.11 $\pm$ 0.03	5.47 $\pm$ 0.78	0.92(0.8–0.97)	0.95(0.89–0.98)
CF-Loss $Loss_{CF-V}$	71.32 $\pm$ 1.89	72.72 $\pm$ 0.81	57.3 $\pm$ 0.99	2.09 $\pm$ 0.06	5.8 $\pm$ 0.72	0.89(0.73–0.96)	0.91(0.79–0.97)
CF-Loss $Loss_{CF-VB}$	67.7 $\pm$ 1.48	71.59 $\pm$ 1.32	55.96 $\pm$ 1.58	2.08 $\pm$ 0.06	5.18 $\pm$ 1.41	0.9(0.75–0.96)	0.89(0.74–0.96)
p-value	5.38e–3	1.36e–3	3.88e–3	1.36e–3	1.95e–3	N/A	N/A
Evaluation category: vein							
DRIVE-AV							
Loss	Sensitivity $\uparrow$	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _F (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	70.14 $\pm$ 2.49	70.17 $\pm$ 2.43	55.61 $\pm$ 1.76	3.65 $\pm$ 0.23	16.23 $\pm$ 2.38	0.61(0.22–0.81)	0.53(0.14–0.82)
GC (Zheng et al., 2021)	72.95 $\pm$ 1.42	71.58 $\pm$ 2.25	55.73 $\pm$ 1.38	3.57 $\pm$ 0.14	15.17 $\pm$ 2.93	0.63(0.28–0.84)	0.62(0.23–0.78)
clDice (Shit et al., 2021)	74.66 $\pm$ 2.39	73.41 $\pm$ 1.25	57.68 $\pm$ 1.32	3.52 $\pm$ 0.15	13.21 $\pm$ 2.29	0.65(0.31–0.87)	0.62(0.18–0.85)
CF-Loss $Loss_{CF-B}$	78.98 $\pm$ 2.48	76.44 $\pm$ 0.4	61.98 $\pm$ 0.53	3.16 $\pm$ 0.09	10.98 $\pm$ 2.77	0.82(0.57–0.92)	0.71(0.48–0.88)
CF-Loss $Loss_{CF-V}$	77.61 $\pm$ 1.36	76.33 $\pm$ 0.54	61.83 $\pm$ 0.72	3.13 $\pm$ 0.12	11.14 $\pm$ 2.06	0.79(0.49–0.92)	0.68(0.29–0.87)
CF-Loss $Loss_{CF-VB}$	77.21 $\pm$ 2.31	76.56 $\pm$ 0.95	62.15 $\pm$ 1.24	3.07 $\pm$ 0.15	9.45 $\pm$ 1.95	0.8(0.49–0.92)	0.69(0.34–0.85)
p-value	1.36e–3	9.39e–4	9.39e–4	1.36e–2	1.36e–2	N/A	N/A
LES-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _F (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	63.43 $\pm$ 2.36	68.19 $\pm$ 1.97	53.34 $\pm$ 1.88	2.85 $\pm$ 0.14	8.23 $\pm$ 2.38	0.72(0.44–0.88)	0.74(0.48–0.91)
GC (Zheng et al., 2021)	62.13 $\pm$ 2.12	67.98 $\pm$ 1.78	53.21 $\pm$ 1.73	2.76 $\pm$ 0.17	6.29 $\pm$ 2.06	0.85(0.76–0.92)	0.84(0.73–0.91)
clDice (Shit et al., 2021)	64.18 $\pm$ 1.91	69.34 $\pm$ 1.45	54.54 $\pm$ 1.89	2.68 $\pm$ 0.08	4.45 $\pm$ 0.94	0.87(0.78–0.94)	0.86(0.76–0.93)
CF-Loss $Loss_{CF-B}$	67.42 $\pm$ 2.73	71.96 $\pm$ 1.87	57.46 $\pm$ 1.34	2.4 $\pm$ 0.11	2.73 $\pm$ 0.74	0.98(0.93–0.99)	0.96(0.85–0.99)
CF-Loss $Loss_{CF-V}$	66.29 $\pm$ 2.49	71.19 $\pm$ 1.69	56.75 $\pm$ 1.78	2.42 $\pm$ 0.09	2.59 $\pm$ 0.45	0.94(0.78–0.98)	0.92(0.73–0.98)
CF-Loss $Loss_{CF-VB}$	68.39 $\pm$ 1.16	72.29 $\pm$ 0.89	57.78 $\pm$ 0.82	2.39 $\pm$ 0.07	2.66 $\pm$ 0.5	0.91(0.66–0.98)	0.91(0.68–0.98)
p-value	2.76e–3	1.95e–3	1.95e–3	9.39e–4	3.88e–3	N/A	N/A
HRF-AV							
Loss	Sensitivity	F1-score $\uparrow$	IOU $\uparrow$	MSE $\downarrow$	Betti error $\downarrow$	ICC _A (95%CI) $\uparrow$	ICC _V (95%CI) $\uparrow$
AC (Chen et al., 2019)	71.51 $\pm$ 2.25	72.36 $\pm$ 1.36	58.24 $\pm$ 0.97	2.36 $\pm$ 0.04	12.03 $\pm$ 1.39	0.69(0.33–0.84)	0.71(0.22–0.85)
GC (Zheng et al., 2021)	71.17 $\pm$ 2.48	72.36 $\pm$ 1.58	59.27 $\pm$ 1.15	2.2 $\pm$ 0.05	15.02 $\pm$ 2.12	0.68(0.28–0.87)	0.69(0.42–0.84)
clDice (Shit et al., 2021)	71.26 $\pm$ 2.64	73.54 $\pm$ 0.83	59.19 $\pm$ 1.32	2.2 $\pm$ 0.05	11.49 $\pm$ 1.7	0.72(0.41–0.89)	0.71(0.36–0.92)
CF-Loss $Loss_{CF-B}$	74.14 $\pm$ 2.26	76.89 $\pm$ 0.59	62.57 $\pm$ 0.77	1.99 $\pm$ 0.03	8.91 $\pm$ 0.82	0.77(0.44–0.91)	0.82(0.55–0.93)
CF-Loss $Loss_{CF-V}$	77.79 $\pm$ 2.86	77.61 $\pm$ 0.9	63.53 $\pm$ 1.16	2.02 $\pm$ 0.05	10.27 $\pm$ 1.55	0.72(0.31–0.89)	0.72(0.32–0.89)
CF-Loss $Loss_{CF-VB}$	75.58 $\pm$ 2.63	77.1 $\pm$ 0.59	62.86 $\pm$ 0.77	2.01 $\pm$ 0.07	9.72 $\pm$ 2.18	0.67(0.29–0.87)	0.71(0.3–0.89)
p-value	4.06e–2	9.39e–4	1.36e–3	9.39e–4	2.76e–3	N/A	N/A

functions. This will be studied in future statistical research. Additionally, we believe that embedding medical and physical prior knowledge of vessel characteristics, morphology and bifurcation in deep learning models may further benefit multi-class vessel segmentation. The prior knowledge of morphological feature distribution observed on large-scale data, such as volume prior in cardiac imaging (Kervadec et al., 2019), can work as effective regularisations for deep learning model training. While machine learning-based models are more sensitive to the difference of pixel density between several categories, physicians normally have much stronger logic in distinguishing multi-class vessels. For example, physicians can infer the category of some challenging

downstream vessel branches based on the knowledge of upstream vessels, which is hardly achieved even by state-of-the-art methods. The formulation and integration of medical and physical prior knowledge in model development will boost the accuracy of feature measurement and multi-class segmentation.

More broadly, our results motivate a wider use of feature-based loss functions in medical image computing tasks, as feature measurement provides a quantitative and interpretable reference for disease diagnosis, progression monitoring, and disease marker discovery in various medical fields, such as the anatomical volumes in the brains, lesion load in the prostate, and fibrosis density in the lungs. By building

the feature-based loss function to regulate the model training, more accurate clinically-relevant features can be obtained compared to using pixel-based loss functions alone.

CF-Loss contributes to downstream clinical research, such as ophthalmics, and potentially promotes the deployment of automated AI techniques in clinical applications. Beyond an observation of visible pathologies, such as exudates and haemorrhages, vascular features can reveal imperceptible morphology alteration which potentially implies circulatory and metabolic dysfunction. By integrating our method into the clinical daily routine, both vascular features and multi-class vessel maps can be supplied as an auxiliary tool to dig implicit disease-related information. To this end, further future work will evaluate the robustness of CF-Loss on large-scale clinical datasets, such as AlzEye data, and a wider range of disease diagnosis tasks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data for multi-class vessel segmentation is publicly available. Data for clinical experiment is not allowed to share.

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